

Data Quality Report

RecoMart Recommendation Pipeline | Generated: July 21, 2026

Overall Data Quality Score: 100.0% (10/10 checks passed)

1. Product Data Validation

Dataset: Kaggle Ecommerce Product Recommendation Dataset

Raw records: 1,474 | After deduplication: 1,018 | 456 duplicates removed

Check	Status	Details
Schema Check	PASS	All 13 expected columns present
Missing Values	PASS	No critical missing values above threshold
Duplicate Rows	PASS	456 duplicates detected and removed. 1,018 clean rows retained.
Rating Range	PASS	All ratings within valid range (0.0 - 5.0)
Price Range	PASS	All prices are positive (> 0)
Data Types	PASS	All columns match expected data types

2. User Interaction Data Validation

Dataset: Simulated REST API User Interactions

Records: 5,000 | Users: 200 | Action types: view, click, purchase, rate

Check	Status	Details
Schema Check	PASS	All 5 expected columns present (user_id, product_id, rating, timestamp, action_type)
Rating Range	PASS	All explicit ratings within 1.0 - 5.0
Valid Action Types	PASS	All action_type values in {view, click, purchase, rate}
Timestamp Format	PASS	All timestamps parseable as valid datetime

3. Data Quality Issue Handling

3.1 Duplicate Records (456 rows)

Root Cause: The Kaggle dataset contained 1,474 raw rows with 456 exact duplicate records. This is likely due to repeated product listings across different campaigns or data export artefacts.

Resolution: The validation pipeline automatically detects and removes all exact duplicate rows BEFORE running quality checks. The validated CSV contains only 1,018 unique product records. The duplicate check is marked PASS to reflect that the issue was actively resolved, not just flagged.

3.2 Missing Values

No critical missing values were detected above the configured threshold. Any remaining minor nulls are handled

downstream by the data preparation stage using median imputation for numerical columns and mode imputation for categorical columns.

4. Summary Statistics

Metric	Product Data	Interaction Data
Raw Records	1,474	5,000
After Deduplication	1,018	5,000
Columns	13	5
Missing Values (critical)	0	0
Validation Checks	6 / 6 PASS	4 / 4 PASS
Overall Quality Score	100%	100%

5. Recommendations

- Schedule periodic deduplication checks after each Kaggle dataset update
- Consider adding outlier detection for extreme price values in future iterations
- Expand rating interaction volume to improve SVD collaborative filtering performance
- Add schema versioning to detect upstream data drift automatically